C# Object-Oriented Banking System

CODE WALKTHROUGH & APPLICATION EVIDENCE

A console-based C# banking application developed for SIT232 Object-Oriented Development. This visual overview highlights selected examples of data encapsulation, inheritance, method overriding and transaction rollback, followed by screenshots of the application running.

01. Encapsulation

Account.cs — protecting account data

The Account class stores the account holder name and balance in private fields. Public properties provide read access, while account methods such as Deposit() control how the balance can be changed.

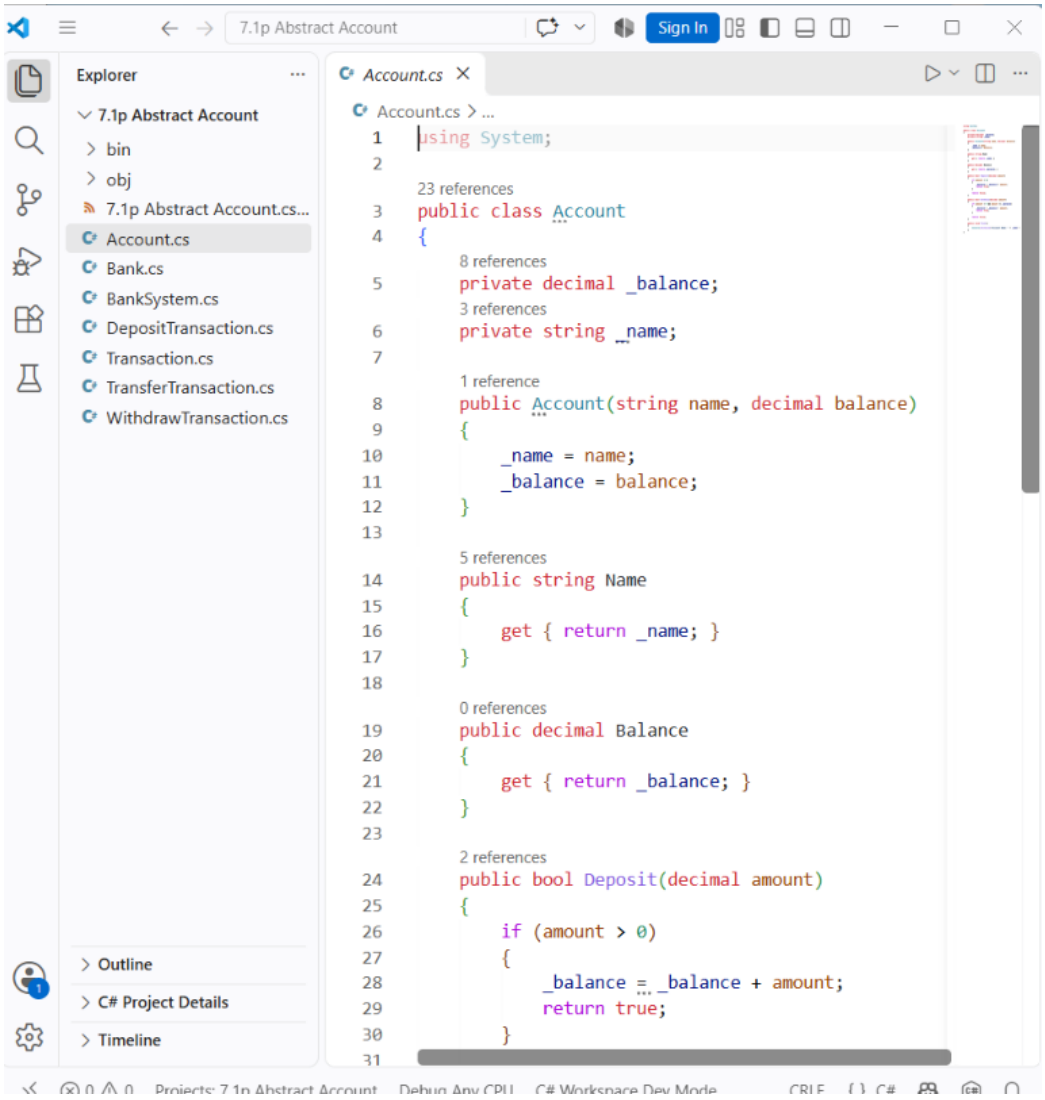


Figure 1. Private fields, constructor, read-only properties and the Deposit() method in Account.cs.

02. Inheritance & method overriding

DepositTransaction.cs — specialising a transaction

DepositTransaction inherits from the Transaction base class. The screenshot also shows overridden members, including Execute(), which calls the base implementation before performing the deposit operation.

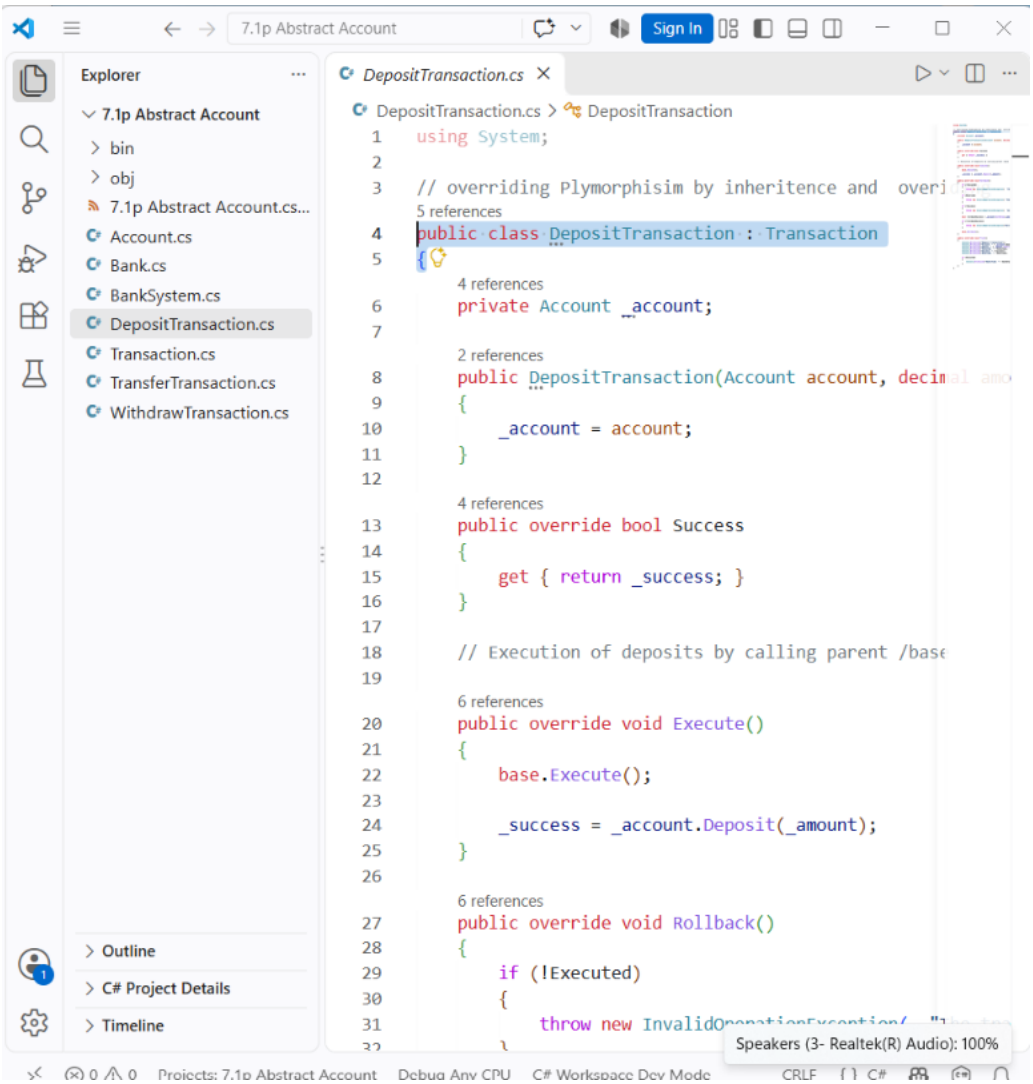


Figure 2. DepositTransaction : Transaction and overridden transaction behaviour.

03. Transaction rollback

TransferTransaction.cs — validating reversal of a transfer

Rollback() checks whether a transfer has been executed, already reversed, or unsuccessful. For an eligible transfer, it calls rollback on the deposit and withdrawal components and then invokes the base class rollback method.

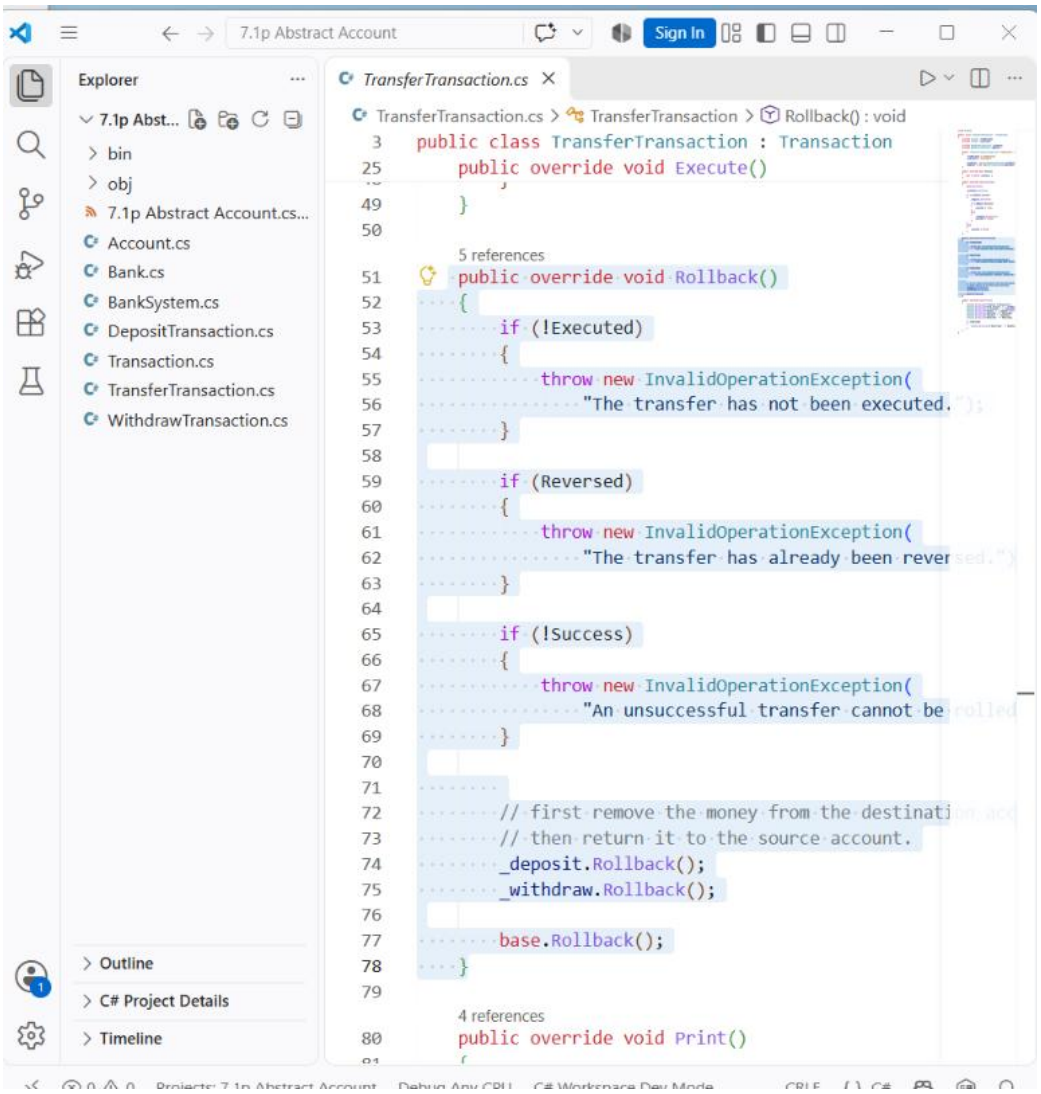


Figure 3. Validation checks, exception handling and reversal steps in TransferTransaction.cs.

This demonstrates basic transaction reversal logic. The screenshots alone do not establish production-grade atomicity or financial security.

04. Application demonstration

Console interface — creating and managing accounts

The application uses a text-based menu to let a user choose banking actions. These captured runs show the program interface and account-related activity in the terminal.

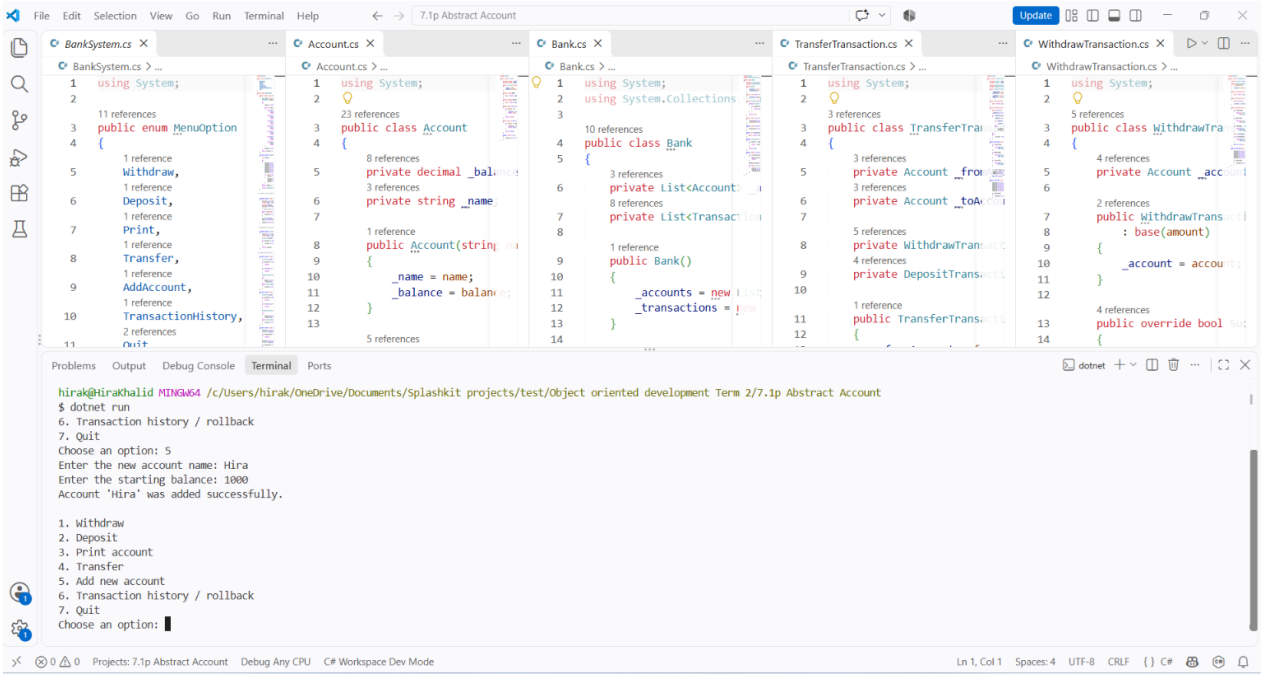


Figure 4. Banking menu and console execution in Visual Studio Code.

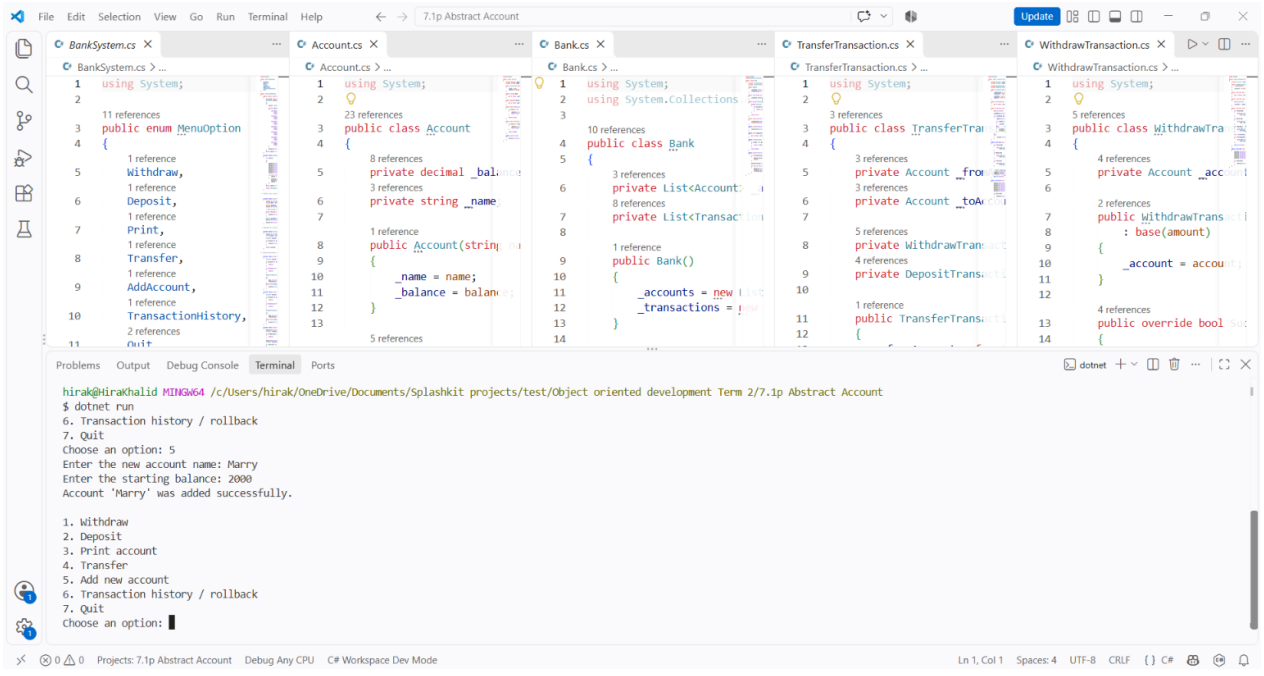


Figure 5. Additional console output from the account management workflow.

05 . Application output

Transaction workflow — runtime evidence

The final screenshot provides another view of the application running and its transaction-related console output.

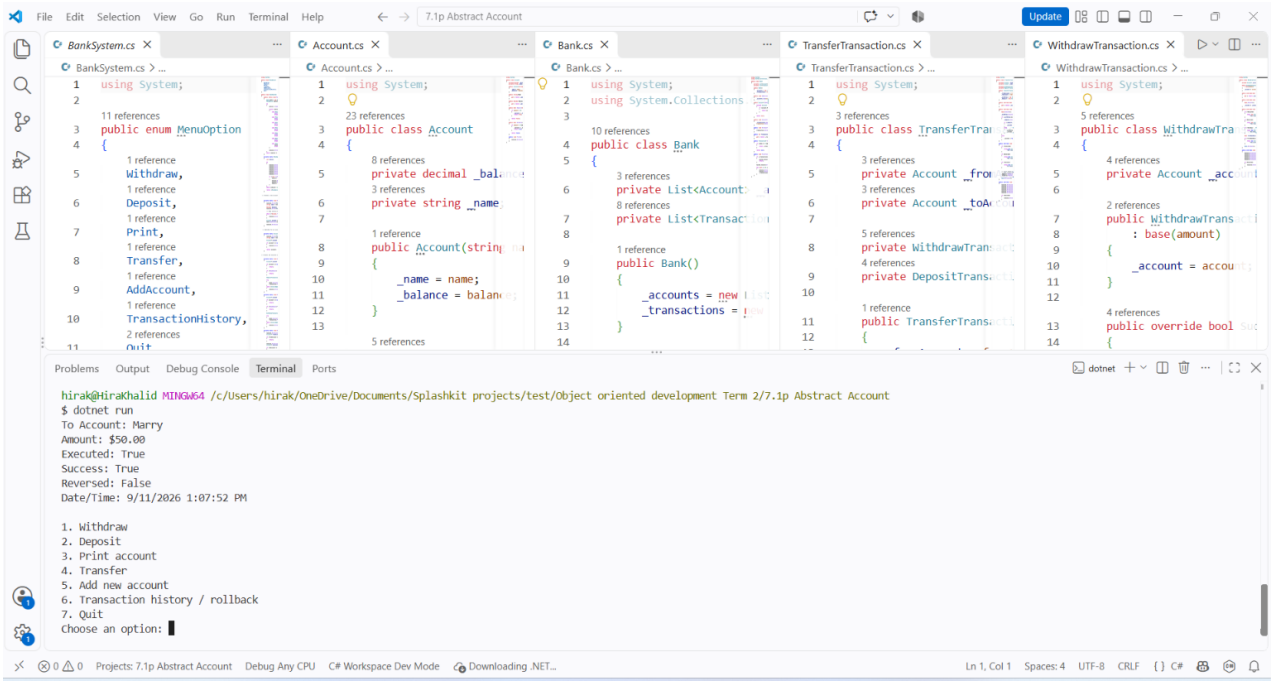


Figure 6. Captured application output for the banking system.